

Ariffin Samsu

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PROFESSIONAL SUMMARY

Results-driven professional with 3 years of experience in high-stakes, data-intensive environments — semiconductor manufacturing (GlobalFoundries, Singapore) and asset integrity management (DNV Malaysia). Proficient in using Excel (PivotTable, XLOOKUP, advanced functions), Power Query, and Power BI to transform raw operational data into actionable insights, build dashboards, and track KPIs that drive measurable business outcomes: reduced defects by up to 21%, cut scrap costs by 15%, and delivered ~SGD 250,000 in annual savings through data-backed decisions. Strong foundation in statistical analysis, root cause analysis, and continuous improvement methodologies such as Lean Six Sigma, DMAIC, and DOE. Experienced in collaborating with cross-functional teams to improve process performance, monitor trends, and support data-driven decisions. Currently targeting Junior Data Analyst roles in Kuala Lumpur.

CORE SKILLS

Analytics & BI Tools: Power BI (dashboard development, KPI reporting) | Microsoft Excel (PivotTables, VLOOKUP, data validation) | JMP

Data & Querying: SQL (MySQL) | Statistical Process Control (SPC) | Fault Detection & Classification (FDC) | ETL workflows | Data cleaning & validation

Programming: Python (pandas, matplotlib)

Methodologies: Root Cause Analysis (RCA) | 8D | DMAIC | Lean Six Sigma Yellow

Soft Skills: Cross-functional stakeholder communication | Data storytelling | Process documentation | Multicultural team collaboration

PROFESSIONAL EXPERIENCE

DNV Malaysia

Nov 2024 – Feb 2026

Asset Integrity Management (AIM) Engineer

- Analysed operational and inspection data across a portfolio of 700+ assets (pressure vessels, piping systems, storage tanks) to identify failure risk scenarios and prioritise maintenance actions — directly analogous to data-driven prioritisation in a business context.
- Interpreted degradation and NDT inspection data for 93 equipment units, translating technical findings into forward-planning recommendations for maintenance and resource allocation decisions.
- Built risk-based assessment models to evaluate asset condition, enabling scenario-based planning (repair vs. replace vs. monitor) — applying structured analytical frameworks equivalent to commercial data analysis.
- Strengthened cross-functional stakeholder communication by presenting data-driven risk findings to engineers, project leads, and client representatives in clear, actionable formats.

Note: Role concluded at contract end due to project cycle slowdown — not performance-related.

GlobalFoundries – Fab7, Singapore

Oct 2022 – Aug 2024

Process Engineer, STI CMP | Data Analytics & Performance Reporting

- Built and maintained Power BI dashboards tracking daily department KPIs — tool performance, process conditions, layer thickness outputs, and defect scan results — enabling real-time visibility for team leads and shift managers.
- Transformed raw SPC and FDC data streams into actionable process insights, identifying variability patterns and risk signals that resulted in a 10% improvement in process stability and a 12% reduction in product-level defects.
- Reduced monthly scrap rate by 15% by conducting structured root cause analysis on defect data and implementing targeted process corrections — contributing directly to inventory cost optimisation.
- Delivered process improvements that reduced particle-related defects by 21% and lowered tool energy consumption by 8%, supporting both operational efficiency and sustainability KPIs.
- Led greenfield tool and product qualifications end-to-end, independently analysing qualification data against defect density and throughput benchmarks to validate readiness for high-volume manufacturing.
- Identified and executed supplier optimisation opportunities through cost-benefit data analysis, delivering ~SGD 250,000 in annual savings while maintaining production performance targets.

- Mentored 2 junior engineers in data interpretation, process control methodologies, and SPC-based troubleshooting.

Top Glove – F18, Malaysia

Aug 2021 – Oct 2021

Trainee (Production Intern)

- Led a cost-reduction analysis project, identifying an opportunity to reduce alkaline mass input by 33% while maintaining product quality — validated through controlled data collection and process monitoring.
- Applied CFD chemical flow simulation to model fluid circulation patterns and propose tank homogeneity improvements.

EDUCATION

Bachelor of Chemical Engineering (Hons)

2018 – 2022

Universiti Sains Malaysia | CGPA: 3.61

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Google Data Analytics Professional Certificate — Coursera (2025)
- Lean Six Sigma Yellow Belt
- LinkedIn Learning: Become a Data Analyst path

LANGUAGES

- Malay — Native
- English — Professionally Fluent